DEGREES OF PROLONGATION VARIETIES OF MONOMIAL PLANAR SYSTEMS

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ABSTRACT. For the differential system $x_1' = x_2^p$, $x_2' = x_1^q$ with $p, q \geq 2$ we compute the degree of every prolongation variety. It is a quadratic quasi-polynomial in the order of prolongation, with period $\operatorname{lcm}(p+1, q+1)$. This proves the two formulas proposed by Pogudin, gives the quasi-polynomial he asked for when $(p, q) = (2, 3)$, and proves Puddu's conjecture.

1. INTRODUCTION

Let K be an algebraically closed field of characteristic zero, regarded as a differential field with the zero derivation, and let $f_1, \ldots, f_\ell$ be differential polynomials in $x_1, \ldots, x_n$. The m-th prolongation of the system $f_1 = \cdots = f_\ell = 0$ is the affine variety

$$X_m = V\big(f_i^{(h)} : 1 \leq i \leq \ell,\ 0 \leq h \leq m\big),$$

cut out by the equations and their first m derivatives in the finitely many jet coordinates that occur [13, 11]. Prolongations are the finite-dimensional shadows of the differential system, and their dimension, components and degree govern the cost of methods that truncate a differential problem at a finite order. Pogudin [11, Section 2] singled out the case of a polynomial dynamical system $x' = P(x)$. Here X_m is irreducible of dimension n for every m, so the only invariant left to determine is its degree.[1]

Already for $n = 2$ the answer is not obvious. For the systems

$$F_1 :\ x_1' = x_2^2,\ x_2' = x_1^2, \qquad F_2 :\ x_1' = x_2^3,\ x_2' = x_1^3,$$

Pogudin computed $\deg X_m$ for $m \leq 10$ and proposed

$$\deg X_m(F_1) = (m+2)^2 - 2\,[\,m \equiv 1\ (3)\,], \qquad \deg X_m(F_2) = (2m+3)^2 - 8\,[\,m \equiv 1, 2\ (4)\,], \tag{1.1}$$

asking for a proof (Question 6 of [11]). He also asked whether $\deg X_m$ is always a quasi-polynomial in m and how it depends on the system (Question 7), and, for

$$F_3 :\ x_1' = x_2^2,\ x_2' = x_1^3,$$

listed the values $6, 11, 26, 46, 66, 91, 121, 156, 196$ and asked what the quasi-polynomial would be. Upper bounds for F_1 were found by Jeronimo and by Falkensteiner and Sendra.[2] Lanciano [10] proved $\deg X_1 \leq (\deg X_0)^2$ for systems with a smooth irreducible zero set and a prime defining ideal, which gives 16 for F_1 against the true value 7. Puddu [12] proved the first formula in

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[1] The degree of an irreducible affine variety of dimension d is the number of points in which a generic affine linear subspace of codimension d meets it [9, Lecture 18]. For X_m it depends on the embedding by jet coordinates, not only on the abstract variety.

[2] Both are reported in [11, Section 2] as personal communications.

(1.1) with algebraic Thomas decompositions and an interlacing argument, and conjectured for $x_1' = x_2^n$, $x_2' = x_1^n$ the formula $\deg X_m = ((n-1)m+n)^2 - 2(n-1)r(n-r)$, where $m \equiv r \pmod{n+1}$ [12, Conjecture 4.1].

We settle these questions for the whole two-parameter family

$$F_{p,q}: \qquad x_1' = x_2^p, \qquad x_2' = x_1^q \qquad (p, q \geq 2).$$

Theorem 1. *Let $p, q \geq 2$, put $c = pq - 1$, and for $m \geq 0$ let $r \in \{0, \dots, p\}$ and $s \in \{0, \dots, q\}$ be the residues of m modulo $p+1$ and $q+1$. Then*

$$\deg X_m(F_{p,q}) = \frac{(p(q+1)+mc)(q(p+1)+mc)}{(p+1)(q+1)} - c\left(\frac{r(p-r)}{p+1} + \frac{s(q-s)}{q+1}\right).$$

The first term is a weighted Bézout number and the bracket records two points at infinity, one on each coordinate axis, at which the generic linear section of X_m escapes to infinity; Figure 1 shows the bracket for $(p, q) = (3, 4)$. For $m = 0$ the formula gives pq, the degree of the graph of (x_2^p, x_1^q).

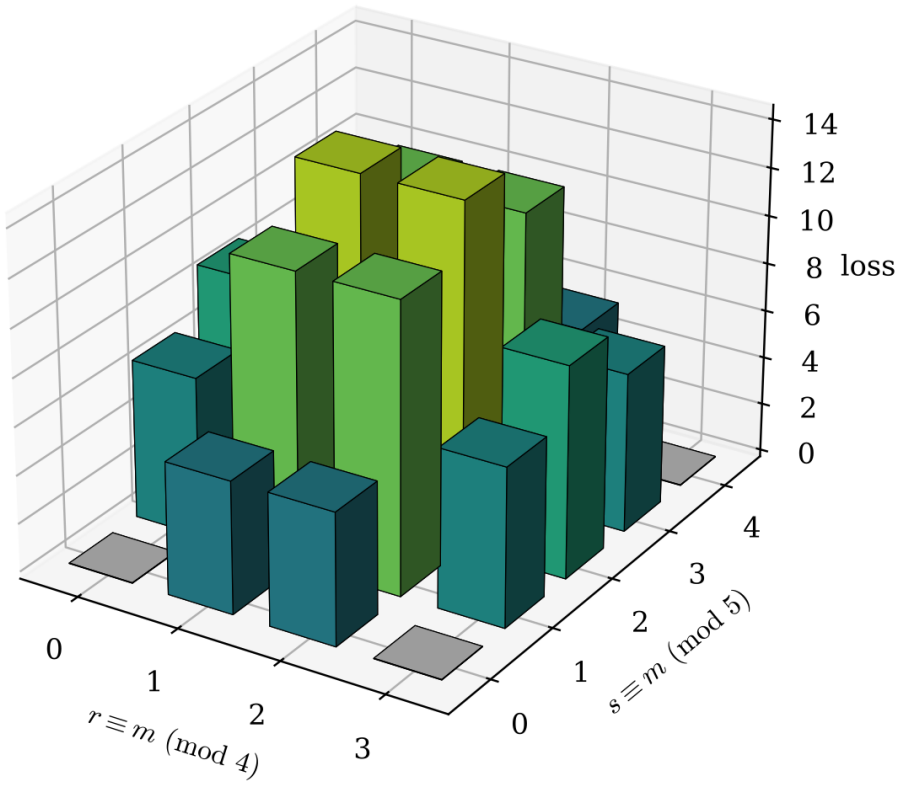


FIGURE 1. The loss $c(r(p-r)/(p+1) + s(q-s)/(q+1))$ in Theorem 1 for $(p, q) = (3, 4)$, $c = 11$, over the residues r of m modulo 4 and s modulo 5. The bars on the edges $r \in \{0, p\}$ and $s \in \{0, q\}$ come from one point at infinity only; the loss is zero exactly when $r \in \{0, p\}$ and $s \in \{0, q\}$.

Corollary 2. (i) *Both formulas* (1.1) *hold for every* $m \geq 0$. *More generally, for* $p = q = n \geq 2$ *one has* $\deg X_m = ((n-1)m+n)^2 - 2(n-1)r(n-r)$ *with* $r \equiv m \pmod{n+1}$, *which is Puddu's conjecture.*

(ii) *For* F_3, *with* $r \equiv m \pmod 3$ *and* $s \equiv m \pmod 4$,

$$\deg X_m(F_3) = \frac{(5m+8)(5m+9) - 20\,r(2-r) - 15\,s(3-s)}{12},$$

which gives $6, 11, 26, 46, 66, 91, 121, 156, 196, 236, 281, \ldots$ *for* $m = 0, 1, 2, \ldots$.

(iii) *For every* $p, q \geq 2$ *the sequence* $\deg X_m(F_{p,q})$ *is a quasi-polynomial of degree two in* m, *with leading coefficient* $(pq-1)^2/((p+1)(q+1))$ *and periodic part of period* $\operatorname{lcm}(p+1, q+1)$.

The prolongation X_m is the graph of the map $(a,b) \mapsto (A_l(a,b), B_l(a,b))_{1 \leq l \leq m+1}$, where A_l and B_l are the iterates of a and b under the derivation $D = b^p\partial_a + a^q\partial_b$ (Lemma 2.1). The second ingredient of the proof, which is of independent interest, describes their common factor; for $p = q$ it is the gcd statement of [12, Conjecture 4.1].

Theorem 3. *Let* $p, q \geq 1$ *and* $l \geq 1$, *and let* $r \in \{0, \ldots, p\}$, $s \in \{0, \ldots, q\}$ *be the residues of* $l-1$ *modulo* $p+1$ *and* $q+1$. *Every common zero of* A_l *and* B_l *in* K^2 *lies on the coordinate axes, and*

$$\gcd(A_l, B_l) = a^{(q-s)[\,s \geq 1\,]}\, b^{(p-r)[\,r \geq 1\,]}.$$

Method. Theorem 1 is proved in three steps. First, D is homogeneous for the weights $\operatorname{wt} a = (p+1)/g$, $\operatorname{wt} b = (q+1)/g$ with $g = \gcd(p+1, q+1)$, and a generic linear section of X_m is the zero set of $A_{m+1} + U$ and $B_{m+1} + V$ with U, V of lower weight. In the weighted projective plane the two curves meet in a number of points given by their Hilbert series, and the affine count is that number minus the intersection multiplicities at infinity (Section 4). Second, Theorem 3 shows that the only possible points at infinity are the two axis points. We prove it by transporting A_l and B_l to a real arc of the curve $u^{p+1} + v^{q+1} = 1$, where they become successive derivatives along a flow, and by showing that their zeros on the arc interlace at every order (Section 3). Third, at each axis point both curves have a Newton polygon with a single edge, which gives the multiplicities $c\,r(p-r)/g$ and $c\,s(q-s)/g$ (Section 5).

For $p = q$ the jets are homogeneous, the two points at infinity are exchanged by the symmetry $a \leftrightarrow b$, and the real arc is the curve of the squigonometric functions of van Lith [14].[3] Puddu indicated in [12, Section 4] that the multiplicity at each axis point should be $(n-1)r(n-r)$, and a proof outline for $p = q$ along these lines was posted on 5 November 2026 in a public issue of a formalisation project.[4] When $p \neq q$ the jets are only weighted homogeneous, the two axis points carry different multiplicities, and the relation between A_l and B_l on the arc is no longer a reflection; Sections 4–5 treat the general case directly and contain the case $p = q$.

Exact formulas that pin down, for a whole family, a count first known from computation are also the subject of the author's earlier work on Lehmer's totient problem [2], on class groups of small exponent [3] and on Tate–Shafarevich groups [1].

Organisation. Section 2 sets up the jets and their weights. Section 3 proves Theorem 3. Section 4 reduces the degree to two local intersection numbers, Section 5 computes them, and Section 6 completes the proofs. No step relies on a computer; the values in Corollary 2(ii) agree with those listed in [11].

[3] For $p = q = n$ the flow of Section 3, rescaled by $n+1$, is $u' = v^n$, $v' = -u^n$ on $u^{n+1} + v^{n+1} = 1$; for general p, q the arc is of the kind studied for generalised trigonometric functions [7].

[4] Issue #26996 of the repository `https://github.com/the-omega-institute/trureturing`, which records the outline and no formal proof.

2. The jets and their weights

Throughout, $p, q \geq 1$ are integers, $N = p+1$, $M = q+1$, and for an integer t and a modulus k we write $t \bmod k$ for the residue in $\{0, \dots, k-1\}$. The bracket $[\,P\,]$ is 1 if P holds and 0 otherwise. On $K[a,b]$ let

$$D = b^p \frac{\partial}{\partial a} + a^q \frac{\partial}{\partial b}, \qquad A_l = D^l(a), \qquad B_l = D^l(b) \qquad (l \geq 0).$$

Lemma 2.1. *Let $f_1 = x_1' - x_2^p$ and $f_2 = x_2' - x_1^q$. For every $m \geq 0$ the ideal generated by $f_i^{(h)}$, $i = 1, 2$, $0 \leq h \leq m$, equals the ideal generated by $x_1^{(l)} - A_l(x_1, x_2)$ and $x_2^{(l)} - B_l(x_1, x_2)$, $1 \leq l \leq m+1$. Hence $X_m \subset \mathbb{A}^{2(m+2)}$ is the graph of $(a,b) \mapsto (A_l(a,b), B_l(a,b))_{1 \leq l \leq m+1}$, and $X_m \cong \mathbb{A}^2$.*

Proof. Write I_h for the ideal generated by $f_i^{(k)}$, $k \leq h$, and J_h for the ideal generated by $x_1^{(l)} - A_l$ and $x_2^{(l)} - B_l$, $1 \leq l \leq h+1$; thus $J_0 = I_0 = (f_1, f_2)$. For $g \in K[x_1, x_2]$ the chain rule gives $g' = g_{x_1} x_1' + g_{x_2} x_2' \equiv (Dg)(x_1, x_2)$ modulo J_0. Applied to the generators of J_{h-1}, this shows $J_{h-1}' \subseteq J_h$, since $(x_1^{(l)} - A_l)' \equiv x_1^{(l+1)} - A_{l+1}$ modulo J_0, and similarly for x_2. Applied repeatedly to g, it gives $g^{(h)} \equiv (D^h g)(x_1, x_2)$ modulo J_{h-1}. With $g = x_2^p$ and $g = x_1^q$, and since $A_{h+1} = D^h(b^p)$ and $B_{h+1} = D^h(a^q)$, we get

$$f_1^{(h)} \equiv x_1^{(h+1)} - A_{h+1}, \qquad f_2^{(h)} \equiv x_2^{(h+1)} - B_{h+1} \qquad \text{modulo } J_{h-1}.$$

Hence $I_h \subseteq J_h$, and by induction on h also $J_h \subseteq I_h + J_{h-1} \subseteq I_h$. □

Let $V_m \subset K[a,b]$ be the span of $1, A_0, B_0, \dots, A_m, B_m$.

Lemma 2.2. *There is a dense open set $\Omega \subset V_m \times V_m$ such that for $(U, V) \in \Omega$ the polynomials $F = A_{m+1} + U$, $G = B_{m+1} + V$ satisfy*

$$\deg X_m = \dim_K K[a,b]/(F, G),$$

and every common zero of F and G is a simple one.

Proof. An affine subspace of codimension two is the zero set of two affine linear forms $\lambda_i = c_i + \sum_{l \leq m+1} (u_{il} x_1^{(l)} + v_{il} x_2^{(l)})$. By Lemma 2.1, its intersection with X_m is identified with the zero set in K^2 of $\lambda_i(A_\bullet, B_\bullet)$. If the 2×2 matrix $(u_{i,m+1}, v_{i,m+1})$ is invertible, which is an open condition, a linear change of the two forms, which does not change their zero set, brings them to $F = A_{m+1} + U$ and $G = B_{m+1} + V$ with $U, V \in V_m$, and every pair (U, V) arises. For a generic subspace the intersection consists of $\deg X_m$ points, at each of which it is transversal [9, Lectures 17 and 18]. These conditions depend only on the zero set and define a dense open set of subspaces, so the pairs (U, V) for which they hold form a dense open set Ω. Since $X_m \cong \mathbb{A}^2$ through (a, b), transversality at a point means that dF and dG are independent there, that is, the local ring of $K[a,b]/(F,G)$ at the point is K. □

2.1. Weights.

Let $g = \gcd(N, M)$ and

$$\alpha = \frac{N}{g}, \qquad \beta = \frac{M}{g}, \qquad \delta = \frac{pq-1}{g}, \qquad L = \frac{NM}{g} = N\beta = M\alpha.$$

These are integers ($pq - 1 = (p+1)(q+1) - N - M$), and $p\beta = \alpha + \delta$, $q\alpha = \beta + \delta$. Give a weight α and b weight β. Then D raises weights by δ, so A_l and B_l are weighted homogeneous of weights

$$W_A(l) = \alpha + l\delta, \qquad W_B(l) = \beta + l\delta.$$

For $p, q \geq 2$ we have

$$\delta - |\alpha - \beta| = \frac{pq - 1 - |p - q|}{g} = \frac{(\min(p,q) - 1)(\max(p,q) + 1)}{g} > 0. \tag{2.1}$$

On a monomial, $D(a^i b^j) = i\, a^{i-1} b^{j+p} + j\, a^{i+q} b^{j-1}$. Both terms lower i by one modulo M and j by one modulo N, and both have nonnegative coefficients. Put

$$\rho_l = (-l) \bmod N, \quad \sigma_l = (1-l) \bmod M, \quad \rho'_l = (1-l) \bmod N, \quad \sigma'_l = (-l) \bmod M.$$

Lemma 2.3. *The coefficients of A_l and B_l are nonnegative integers. Every monomial $a^i b^j$ of A_l has $j \equiv \rho_l \pmod N$ and $i \equiv \sigma_l \pmod M$; every monomial of B_l has $j \equiv \rho'_l$ and $i \equiv \sigma'_l$. Moreover A_l contains a monomial with b-exponent exactly ρ_l and one with a-exponent exactly σ_l, and B_l contains a monomial with b-exponent ρ'_l and one with a-exponent σ'_l.*

Proof. The first two statements follow by induction from $A_0 = a$, $B_0 = b$ and the formula for D. Since no cancellation occurs, the monomials of $D(h)$, for h with nonnegative coefficients, are the monomials of $D(a^i b^j)$ over the monomials $a^i b^j$ of h. Let $a^i b^j$ be a monomial of A_l with $j < N$. If $j \geq 1$, then A_{l+1} contains $a^{i+q} b^{j-1}$. If $j = 0$ then $i \geq 1$, since A_l has positive weight, and A_{l+1} contains $a^{i-1} b^p$. In both cases A_{l+1} has a monomial with b-exponent below N, which by the congruence equals ρ_{l+1}. The base case is $A_0 = a$. The other three statements are proved in the same way. □

By Lemma 2.3 the exponents of the monomials of A_l are $j = \rho_l + Nt$ and $i = \sigma_l + Ms$ with $s, t \geq 0$, and weighted homogeneity gives $s + t = k_A(l)$, where

$$L\, k_A(l) = \alpha + l\delta - \sigma_l \alpha - \rho_l \beta, \qquad L\, k_B(l) = \beta + l\delta - \sigma'_l \alpha - \rho'_l \beta. \tag{2.2}$$

Hence

$$A_l = a^{\sigma_l} b^{\rho_l}\, \Pi_l^A(b^N, a^M), \qquad B_l = a^{\sigma'_l} b^{\rho'_l}\, \Pi_l^B(b^N, a^M), \tag{2.3}$$

where $\Pi_l^A(X, Y)$ and $\Pi_l^B(X, Y)$ are binary forms of degrees $k_A(l)$ and $k_B(l)$ with nonnegative integer coefficients, and by the last part of Lemma 2.3 the coefficients of X^k and Y^k in each of them are positive.

Lemma 2.4. *For every $l \geq 0$:*

(a) $k_A(l+1) = k_A(l) - 1 + [\,\rho_l \geq 1\,] + [\,\sigma_l \geq 1\,]$ *and* $k_B(l+1) = k_B(l) - 1 + [\,\rho'_l \geq 1\,] + [\,\sigma'_l \geq 1\,]$;
(b) $k_A(l) + [\,\rho_l = p\,] = k_B(l) + [\,\sigma'_l = q\,]$;
(c) $k_A(0) = k_B(0) = 0$.

Proof. We have $\rho_{l+1} = \rho_l - 1$ if $\rho_l \geq 1$ and $\rho_{l+1} = p$ if $\rho_l = 0$, and similarly for σ. By (2.2), $L(k_A(l+1) - k_A(l)) = \delta - (\sigma_{l+1} - \sigma_l)\alpha - (\rho_{l+1} - \rho_l)\beta$. If $\rho_l, \sigma_l \geq 1$ this is $\delta + \alpha + \beta = L$. If $\sigma_l = 0 < \rho_l$ it is $\delta - q\alpha + \beta = 0$; if $\rho_l = 0 < \sigma_l$ it is $\delta + \alpha - p\beta = 0$; if both vanish it is $\delta - q\alpha - p\beta = -L$. This is (a) for A, and B is the same. For (b), note $\rho'_l = \rho_l + 1$ if $\rho_l < p$ and $\rho'_l = 0$ if $\rho_l = p$, while $\sigma_l = \sigma'_l + 1$ if $\sigma'_l < q$ and $\sigma_l = 0$ if $\sigma'_l = q$. Then (2.2) gives

$$L(k_A(l) - k_B(l)) = \alpha - \beta - (\sigma_l - \sigma'_l)\alpha - (\rho_l - \rho'_l)\beta = L\,[\,\sigma'_l = q\,] - L\,[\,\rho_l = p\,].$$

Part (c) holds because $A_0 = a$ and $B_0 = b$. □

3. Interlacing on a real arc

In this section $K = \mathbb{C}$. Since $A_l, B_l \in \mathbb{Z}[a, b]$, Theorem 3 for $\mathbb{C}$ implies it for every algebraically closed field of characteristic zero: the greatest common divisor of two polynomials with rational coefficients does not change under field extension, and a common zero off the axes would give a common factor other than a and b.

3.1. **The arc.** Consider the real polynomial vector field on $\mathbb{R}^2$

$$\mathcal{X} = \frac{v^q}{N}\frac{\partial}{\partial u} - \frac{u^p}{M}\frac{\partial}{\partial v},$$

and put $\Phi_l = \mathcal{X}^l(v)$ and $\Psi_l = \mathcal{X}^l(u)$. The function $u^N + v^M$ is a first integral, and $\mathcal{X}$ moves the arc

$$\Gamma = \{(u,v) \in (0,1)^2 : \ u^N + v^M = 1\}$$

from $(0,1)$ towards $(1,0)$.

Lemma 3.1. *Choose $\kappa \in \mathbb{C}^\times$ with $\kappa^{pq-1} = -1/(MN^p)$ and put $\mu = N\kappa^q$. The substitution $a = \kappa v$, $b = \mu u$ carries A_l to $\kappa\Phi_l$ and B_l to $\mu\Psi_l$. In particular Φ_l and Ψ_l have the same monomials as A_l and B_l, with $a \leftrightarrow v$ and $b \leftrightarrow u$.*

Proof. Let $\tau : \mathbb{C}[a,b] \to \mathbb{C}[u,v]$ be the substitution. Then $\tau \circ D$ and $\mathcal{X} \circ \tau$ are derivations along τ, and they agree on generators: $\tau(Da) = \mu^p u^p$ and $\mathcal{X}(\tau a) = -\kappa u^p/M$, which are equal because $\mu^p = N^p\kappa^{pq} = -\kappa/M$; and $\tau(Db) = \kappa^q v^q = \mu v^q/N = \mathcal{X}(\tau b)$. Hence $\tau(D^l h) = \mathcal{X}^l(\tau h)$ for all h, and τ maps distinct monomials to nonzero multiples of distinct monomials. □

By Lemma 3.1 and (2.3),

$$\Phi_l = u^{\rho_l} v^{\sigma_l}\, \tilde{\Pi}_l^A(u^N, v^M), \qquad \Psi_l = u^{\rho'_l} v^{\sigma'_l}\, \tilde{\Pi}_l^B(u^N, v^M),$$

where $\tilde{\Pi}_l^A(X,Y) = \kappa^{-1}\kappa^{\sigma_l}\mu^{\rho_l}\,\Pi_l^A(\mu^N X, \kappa^M Y)$ and $\tilde{\Pi}_l^B$ is defined in the same way; these are binary forms of degrees $k_A(l)$ and $k_B(l)$, with real coefficients because Φ_l and Ψ_l are real, and with nonzero coefficients of X^k and Y^k. Parametrise Γ by $x = u^N \in (0,1)$, so that $u = x^{1/N}$ and $v = (1-x)^{1/M}$. Then

$$\varphi_l(x) := \Phi_l|_\Gamma = x^{\rho_l/N}(1-x)^{\sigma_l/M} F_l(x), \qquad \psi_l(x) := \Psi_l|_\Gamma = x^{\rho'_l/N}(1-x)^{\sigma'_l/M} G_l(x), \tag{3.1}$$

where $F_l(x) = \tilde{\Pi}_l^A(x, 1-x)$ and $G_l(x) = \tilde{\Pi}_l^B(x, 1-x)$. Thus $\deg F_l \le k_A(l)$, $\deg G_l \le k_B(l)$, and $F_l(0)F_l(1)G_l(0)G_l(1) \neq 0$.

Lemma 3.2. *Let $\mathcal{U} = x^{p/N}(1-x)^{q/M}\,\dfrac{d}{dx}$ on $(0,1)$. Then $\varphi_{l+1} = \mathcal{U}\varphi_l$ and $\psi_{l+1} = \mathcal{U}\psi_l$.*

Proof. For a polynomial $H(u,v)$, the chain rule gives $\mathcal{U}(H(x^{1/N}, (1-x)^{1/M})) = H_u \cdot \frac{1}{N}(1-x)^{q/M} - H_v \cdot \frac{1}{M}x^{p/N}$, because $(p+1)/N = (q+1)/M = 1$. On Γ this is $(\mathcal{X}H)|_\Gamma$. □

Since the factor $x^{p/N}(1-x)^{q/M}$ is positive on $(0,1)$, Rolle's theorem applies to $\mathcal{U}$ as it does to d/dx.

3.2. **Counting zeros.** For a function θ on $(0,1)$ let $Z(\theta)$ be its set of zeros.

Proposition 3.3. *For every $l \ge 0$ the polynomial F_l has degree $k_A(l)$ and $k_A(l)$ simple zeros, all in $(0,1)$; the same holds for G_l and $k_B(l)$. Moreover, if*

$$E_l = \{0 : \rho_l \ge 1\} \cup Z(\varphi_l) \cup \{1 : \sigma_l \ge 1\} \subset [0,1],$$

then $Z(\varphi_{l+1})$ consists of exactly one point strictly between any two consecutive points of E_l and nothing else. The same holds for ψ_l with $E'_l = \{0 : \rho'_l \ge 1\} \cup Z(\psi_l) \cup \{1 : \sigma'_l \ge 1\}$.

Proof. By (3.1) and $F_l(0)F_l(1) \neq 0$, the function φ_l extends continuously to $[0,1]$, and it vanishes at 0 exactly when $\rho_l \ge 1$ and at 1 exactly when $\sigma_l \ge 1$. Rolle's theorem, applied between consecutive points of E_l, gives

$$|Z(\varphi_{l+1})| \ \ge \ |E_l| - 1 = |Z(\varphi_l)| - 1 + [\,\rho_l \ge 1\,] + [\,\sigma_l \ge 1\,].$$

By Lemma 2.4(a),(c) and induction, $|Z(\varphi_l)| \geq k_A(l)$ for all l. The zeros of φ_l in $(0,1)$ are the zeros of F_l there, and F_l is a nonzero polynomial of degree at most $k_A(l)$. Hence F_l has exactly $k_A(l)$ zeros in $(0,1)$, each simple, and no others, and its degree is $k_A(l)$. The inequalities above are therefore equalities, which gives the statement about E_l. The argument for ψ_l is the same. □

Corollary 3.4. *Every zero of $\tilde{\Pi}_l^A$ in $\mathbb{P}^1(\mathbb{C})$ is of the form $[x : 1-x]$ with $x \in (0,1)$ a zero of F_l, and it is simple. The same holds for $\tilde{\Pi}_l^B$ and G_l.*

Proof. The $k_A(l)$ distinct zeros of F_l in $(0,1)$ give $k_A(l)$ distinct zeros $[x : 1-x]$ of the binary form $\tilde{\Pi}_l^A$, which has degree $k_A(l)$. □

3.3. **Interlacing.** For real polynomials f, g of the same degree $n \geq 0$ we write $f \prec g$ if their zeros are real, simple and satisfy

$$0 \leq \xi_1 < \eta_1 < \xi_2 < \eta_2 < \cdots < \xi_n < \eta_n \leq 1,$$

where ξ_i are the zeros of f and η_i those of g. The following lemma is a variant of the residue argument of [12, Lemma 3.4]; compare [8, Section 1.3].

Lemma 3.5. *Let $f \prec g$ with $\operatorname{lc}(f)\operatorname{lc}(g) > 0$ and put $Q = f/g$. For every $x \in (0,1)$ with $g(x) \neq 0$ and every $D \in (0,1)$,*

$$x(x-1)Q'(x) + D\,Q(x) > 0.$$

Proof. If $n = 0$, Q is a positive constant. Otherwise $Q = Q(\infty) + \sum_i w_i/(x-\eta_i)$ with $w_i = f(\eta_i)/g'(\eta_i)$. Exactly $n-i$ zeros of f and of g' lie to the right of η_i, so $f(\eta_i)$ and $g'(\eta_i)$ have the signs of $(-1)^{n-i}\operatorname{lc}(f)$ and $(-1)^{n-i}\operatorname{lc}(g)$, and $w_i > 0$. As $\eta_i > \xi_1 \geq 0$, we may write

$$Q(x) = Q(0) + \sum_{i=1}^{n} \frac{w_i\, x}{\eta_i(x-\eta_i)},$$

and $Q(0) \geq 0$: $g(0)$ has the sign of $(-1)^n \operatorname{lc}(g)$, while $f(0)$ is zero or has the sign of $(-1)^n \operatorname{lc}(f)$. Since $\frac{d}{dx}\frac{x}{x-\eta} = \frac{-\eta}{(x-\eta)^2}$,

$$x(x-1)Q'(x) + DQ(x) = DQ(0) + \sum_{i=1}^{n} \frac{w_i\, x\,\big((1-x)(1-D)\eta_i + xD(1-\eta_i)\big)}{\eta_i(x-\eta_i)^2}.$$

For $x \in (0,1)$, $D \in (0,1)$ and $\eta_i \in (0,1]$ every summand is positive. □

Put

$$f_l^* = x^{[\rho_l = p]} F_l, \qquad g_l^* = (1-x)^{[\sigma_l' = q]} G_l.$$

By Lemma 2.4(b) and Proposition 3.3 they have the same degree $n_l = k_A(l) + [\,\rho_l = p\,] = k_B(l) + [\,\sigma_l' = q\,]$.

Theorem 3.6. *For every $l \geq 0$, $f_l^* \prec g_l^*$. In particular φ_l and ψ_l have no common zero in $(0,1)$.*

Proof. We argue by induction on l. For $l = 0$, $f_0^* = g_0^* = 1$ and there is nothing to prove. Assume $f_l^* \prec g_l^*$ and let $\tau_l = \operatorname{sgn}(\operatorname{lc}(f_l^*)\operatorname{lc}(g_l^*))$.

Step 1: the Wronskian. From (3.1), using $\rho_l - \rho_l' = -1 + N[\,\rho_l = p\,]$ and $\sigma_l - \sigma_l' = 1 - M[\,\sigma_l' = q\,]$,

$$R_l := \frac{\varphi_l}{\psi_l} = h \cdot \frac{f_l^*}{g_l^*}, \qquad h(x) = x^{-1/N}(1-x)^{1/M}.$$

Since $h'/h = -D_0(x)/(x(1-x))$ with $D_0(x) = (1-x)/N + x/M \in (0,1)$, we obtain, with $Q = f_l^*/g_l^*$,

$$R_l' = -\frac{h}{x(1-x)}\Big(x(x-1)Q' + D_0 Q\Big).$$

Lemma 3.5, applied to (f_l^*, g_l^*) if $\tau_l = 1$ and to $(-f_l^*, g_l^*)$ if $\tau_l = -1$, shows that R_l' has the sign $-\tau_l$ wherever $\psi_l \neq 0$. Let $W_l = \varphi_l'\psi_l - \varphi_l\psi_l'$, so that $W_l = \psi_l^2 R_l'$ off $Z(\psi_l)$. At a zero $\eta \in (0,1)$ of ψ_l we have $W_l(\eta) = -\varphi_l(\eta)\psi_l'(\eta) \neq 0$, because the zero is simple and, by $f_l^* \prec g_l^*$, not a zero of φ_l. By continuity W_l has the constant sign $-\tau_l$ on $(0,1)$.

Step 2: signs at the zeros of ψ_{l+1}. Let $v_1 < \cdots < v_K$ be the zeros of ψ_{l+1} in $(0,1)$, so $K = k_B(l+1)$. As $\psi_l'(v_j) = 0$, Step 1 and Lemma 3.2 give

$$\operatorname{sgn}\varphi_{l+1}(v_j) = -\tau_l \operatorname{sgn}\psi_l(v_j) \neq 0. \tag{3.2}$$

By Proposition 3.3, exactly one point of E_l' lies between v_j and v_{j+1}, and it is a zero of ψ_l at which ψ_l changes sign. So $\psi_l(v_j)$ and $\psi_l(v_{j+1})$ have opposite signs, and by (3.2) φ_{l+1} has a zero in (v_j, v_{j+1}).

Step 3: the left end. Suppose $\rho_l \geq 1$. Near 0, $\varphi_l \sim F_l(0)x^{\rho_l/N}$, so φ_l', and with it φ_{l+1}, has the sign of $\varphi_l(0^+)$ near 0. The sign of $R_l(0^+)$ is τ_l if $\rho_l < p$ (then $f_l^*(0) \neq 0$) and $-\tau_l$ if $\rho_l = p$ (then $\xi_1 = 0$ and f_l^* changes sign at 0). If $1 \leq \rho_l < p$, then $\rho_l' \geq 1$, so $0 \in E_l'$ and ψ_l has no zero in $(0, v_1]$; hence

$$\operatorname{sgn}\varphi_{l+1}(0^+) = \operatorname{sgn}\varphi_l(0^+) = \tau_l \operatorname{sgn}\psi_l(0^+), \qquad \operatorname{sgn}\varphi_{l+1}(v_1) = -\tau_l \operatorname{sgn}\psi_l(0^+).$$

If $\rho_l = p$, then $\rho_l' = 0$, the first point of E_l' is a zero of ψ_l, and ψ_l has exactly one zero in $(0, v_1)$; hence $\operatorname{sgn}\varphi_{l+1}(0^+) = -\tau_l \operatorname{sgn}\psi_l(0^+)$ and $\operatorname{sgn}\varphi_{l+1}(v_1) = \tau_l \operatorname{sgn}\psi_l(0^+)$. In both cases φ_{l+1} has a zero in $(0, v_1)$ when $K \geq 1$.

Step 4: the right end. Suppose $\sigma_l' = 0$, so that $\sigma_l = 1$ and $g_l^* = G_l$ has all its zeros in $(0,1)$. Then $f_l^*(1)$ and $g_l^*(1)$ have the signs of their leading coefficients and $\operatorname{sgn} R_l(1^-) = \tau_l$. Near 1, $\varphi_l \sim F_l(1)(1-x)^{1/M}$, so φ_{l+1} has the sign of $-\varphi_l(1^-)$ there. As $1 \notin E_l'$, the last point of E_l' is a zero of ψ_l and ψ_l has exactly one zero in $(v_K, 1)$. Therefore

$$\operatorname{sgn}\varphi_{l+1}(1^-) = -\tau_l \operatorname{sgn}\psi_l(1^-), \qquad \operatorname{sgn}\varphi_{l+1}(v_K) = -\tau_l \operatorname{sgn}\psi_l(v_K) = \tau_l \operatorname{sgn}\psi_l(1^-),$$

and φ_{l+1} has a zero in $(v_K, 1)$ when $K \geq 1$.

Step 5: counting. Since $\rho_{l+1} = p$ if and only if $\rho_l = 0$, and $\sigma_{l+1}' = q$ if and only if $\sigma_l' = 0$, Lemma 2.4(b) at $l+1$ reads

$$k_A(l+1) = K - 1 + [\,\rho_l \geq 1\,] + [\,\sigma_l' = 0\,].$$

If $K \geq 1$, Steps 2–4 have produced this many distinct zeros of φ_{l+1} in $(0,1)$, none equal to a v_j. By Proposition 3.3 there are no others. So φ_{l+1} has one zero in each (v_j, v_{j+1}), one in $(0, v_1)$ if and only if $\rho_{l+1} < p$, and one in $(v_K, 1)$ if and only if $\sigma_{l+1}' = q$. The zeros of f_{l+1}^* are those of φ_{l+1} together with 0 when $\rho_{l+1} = p$, and the zeros of g_{l+1}^* are the v_j together with 1 when $\sigma_{l+1}' = q$. The list above says exactly that they interlace with a zero of f_{l+1}^* first and a zero of g_{l+1}^* last, that is, $f_{l+1}^* \prec g_{l+1}^*$. If $K = 0$, then $n_{l+1} = [\,\sigma_{l+1}' = q\,] \leq 1$; when $n_{l+1} = 1$ the only zero of g_{l+1}^* is 1 and the only zero of f_{l+1}^* lies in $[0,1)$, so again $f_{l+1}^* \prec g_{l+1}^*$. □

Figure 2 shows the functions φ_l and ψ_l for $(p,q) = (2,3)$.

Proof of Theorem 3. Let $(a_0, b_0) \in \mathbb{C}^2$ be a common zero of A_l and B_l with $a_0 b_0 \neq 0$. By Lemma 3.1, $(u_0, v_0) = (b_0/\mu, a_0/\kappa)$ is a common zero of Φ_l and Ψ_l with $u_0 v_0 \neq 0$, so $[u_0^N : v_0^M]$ is a common zero of $\tilde{\Pi}_l^A$ and $\tilde{\Pi}_l^B$. By Corollary 3.4 it equals $[x_0 : 1-x_0]$ with $x_0 \in (0,1)$ and $F_l(x_0) = G_l(x_0) = 0$, so x_0 is a common zero of φ_l and ψ_l, contrary to Theorem 3.6. Hence every common zero lies on the axes. The greatest common divisor is weighted homogeneous, and

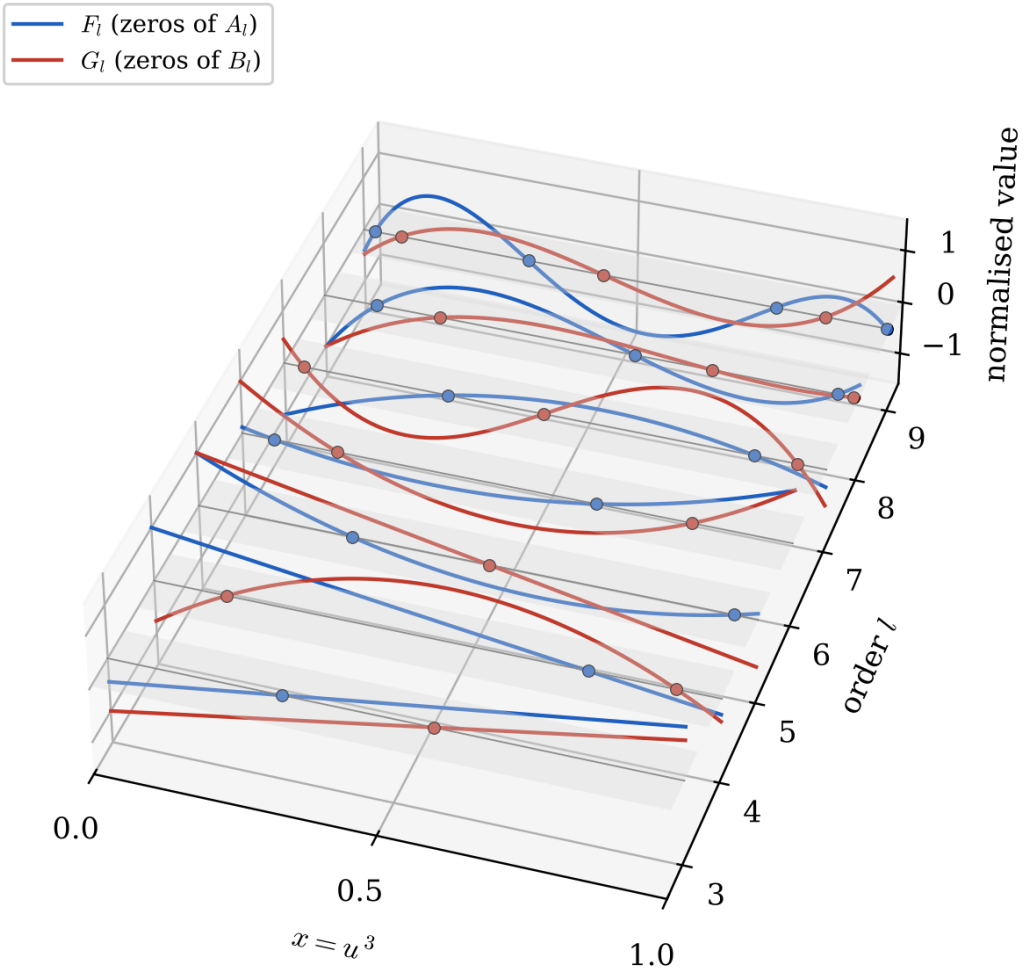


FIGURE 2. The polynomials F_l (blue) and G_l (red) of (3.1) for $(p, q) = (2, 3)$ and $l = 3, \dots, 9$, each scaled to maximum modulus one on $[0, 1]$ and plotted against $x = u^3$. Their zeros, marked by dots, are those of φ_l and ψ_l in $(0, 1)$, and they interlace at every order (Theorem 3.6).

an irreducible factor other than a and b would vanish at a point off the axes. So $\gcd(A_l, B_l) = a^{\min(\sigma_l, \sigma'_l)} b^{\min(\rho_l, \rho'_l)}$ by Lemma 2.3. With $r = (l-1) \bmod N$ we have $\rho_l = p - r$ and $\rho'_l = (p+1-r)[\, r \geq 1\,]$, so the minimum is $(p-r)[\, r \geq 1\,]$; likewise $\min(\sigma_l, \sigma'_l) = (q-s)[\, s \geq 1\,]$. □

4. A weighted Bézout count

From now on $p, q \geq 2$. Fix $m \geq 0$, put $l = m + 1$, and let $F = A_l + U$, $G = B_l + V$ with $(U, V) \in \Omega$ as in Lemma 2.2. By (2.1), every element of V_m has weighted components of weight less than $W_A(l)$ and less than $W_B(l)$: indeed $W_A(j) \leq W_A(l) - \delta$, $W_B(j) \leq W_B(l) - \delta$ and $|W_A(j) - W_B(j)| = |\alpha - \beta| < \delta$ for $j \leq m$. Let $S = K[Z, a, b]$, graded by $\operatorname{wt} Z = 1$, $\operatorname{wt} a = \alpha$, $\operatorname{wt} b = \beta$, and let $F^h, G^h \in S$ be the weighted homogenisations of F, G to weights $W_A = W_A(l)$ and $W_B = W_B(l)$. Then $F^h|_{Z=0} = A_l$ and $G^h|_{Z=0} = B_l$.

For $\lambda \in \{a, b\}$ write ℓ_λ for the intersection multiplicity at the origin of the curves obtained from F^h and G^h by setting $\lambda = 1$, namely

$$\ell_a = \dim_K K[Z, b]_{(Z,b)}/(F^h(Z, 1, b), G^h(Z, 1, b)),$$

and ℓ_b the same with a and b exchanged.

Proposition 4.1. *For (U, V) in a dense open subset of Ω,*

$$\deg X_m = \frac{W_A W_B}{\alpha\beta} - \frac{\ell_a}{\alpha} - \frac{\ell_b}{\beta}.$$

Proof. Regular sequence. A common factor H of F^h and G^h of positive weight is weighted homogeneous and not divisible by Z, because Z divides neither F^h nor G^h. Then $H(1, a, b)$ is a nonconstant common factor of F and G, which is impossible because F and G have finitely many common zeros. Hence F^h, G^h is a regular sequence in S and $M_\bullet = S/(F^h, G^h)$ is a graded Cohen–Macaulay module of dimension one, with Hilbert series

$$\sum_{d\geq 0} \dim_K M_d\, t^d = \frac{(1 - t^{W_A})(1 - t^{W_B})}{(1 - t)(1 - t^\alpha)(1 - t^\beta)}$$

[4, Chapter 4]. Its Hilbert function is eventually periodic, and its average, $\lim_{t\to 1^-}(1 - t)\sum_d \dim M_d t^d$, equals $W_A W_B/(\alpha\beta)$.

Decomposition. Let $T \subset M_\bullet$ be the submodule annihilated by a power of Z and $M' = M_\bullet/T$. The associated primes of M' do not contain Z; they are the ideals $P_x = (a - x_1 Z^\alpha,\, b - x_2 Z^\beta)$ of the common zeros $(x_1, x_2) \in K^2$ of F and G, and $S/P_x \cong K[Z]$. A graded composition series of M' has factors S/Q with $Q \supseteq P_x$ for some x, each finite over $K[Z]$. Hence M' is a finitely generated torsion-free graded $K[Z]$-module, so it is free, and its rank is $\dim_K M'[Z^{-1}]_0 = \dim_K M_\bullet[Z^{-1}]_0 = \dim_K K[a, b]/(F, G) = \deg X_m$ by Lemma 2.2. The Hilbert function of M' is eventually equal to $\deg X_m$.

The torsion. T is supported on $V(Z, A_l, B_l)$, which by Theorem 3 consists of the irrelevant ideal and at most the two lines $P_a = (Z, b)$ and $P_b = (Z, a)$. A graded composition series of T has factors isomorphic to shifts of $S/P_a \cong K[a]$, of $S/P_b \cong K[b]$ and of K; the number of factors S/P_λ is $\operatorname{length}(T_{P_\lambda}) = \operatorname{length}(M_{P_\lambda})$, since no P_x is contained in P_λ. The Hilbert function of a shift of $K[a]$ is eventually periodic with average $1/\alpha$, and that of $K[b]$ with average $1/\beta$. Averaging the Hilbert function of $0 \to T \to M_\bullet \to M' \to 0$ gives

$$\frac{W_A W_B}{\alpha\beta} = \frac{\operatorname{length} M_{P_a}}{\alpha} + \frac{\operatorname{length} M_{P_b}}{\beta} + \deg X_m.$$

The lengths. Let $\mathfrak{m} = (Z, b, a - 1)$. The only minimal prime of the one-dimensional Cohen–Macaulay module $M_\mathfrak{m}$ is P_a, and $a - 1$ is a parameter for it. By the associativity formula for multiplicities [4, Chapter 4],

$$\operatorname{length}(M_\mathfrak{m}/(a - 1)M_\mathfrak{m}) = e(a - 1; M_\mathfrak{m}) = \operatorname{length}(M_{P_a}) \cdot e(a - 1; K[a]_{(a-1)}) = \operatorname{length}(M_{P_a}).$$

The left side is ℓ_a, since $M/(a - 1)M = K[Z, b]/(F^h(Z, 1, b), G^h(Z, 1, b))$. In the same way $\operatorname{length} M_{P_b} = \ell_b$. □

5. The two points at infinity

We keep the notation of Section 4, with $l = m + 1$, and compute ℓ_a. Let $\lambda = l \bmod N$, so that $r = m \bmod N = (\lambda - 1) \bmod N$.

Proposition 5.1. *For (U, V) in a dense open subset of Ω,*

$$\ell_a = (p - r)\, r\, \delta, \qquad \ell_b = (q - s)\, s\, \delta.$$

Proof. Put $f(Z,b) = F^h(Z,1,b)$ and $g(Z,b) = G^h(Z,1,b)$. Each term of f is Z^e times a term b^j of $A_l(1,b)$ ($e=0$), of $A_{l-t}(1,b)$ ($e=t\delta$), of $B_{l-t}(1,b)$ ($e=\alpha-\beta+t\delta$), or of 1 ($e=W_A$), with $1\le t\le l$, and the coefficient of A_{l-t} and B_{l-t} in U multiplies it. The same holds for g with $e=\beta-\alpha+t\delta$ for A_{l-t}, $e=t\delta$ for B_{l-t} and $e=W_B$ for 1. By Lemma 2.3, the b-exponents j of A_{l-t} satisfy $j\equiv t-\lambda$ and those of B_{l-t} satisfy $j\equiv t+1-\lambda \pmod N$.

If $\lambda=0$, then $\rho_l=0$ and $A_l(1,0)\neq 0$, so $f(0,0)\neq 0$ and $\ell_a=0$; if $\lambda=1$, then $\rho'_l=0$ and $g(0,0)\neq 0$. In both cases $r(p-r)=0$. Let now $2\le\lambda\le p$, so $r=\lambda-1\in[1,p-1]$, and put

$$f_P = p+1-\lambda, \quad e_P=\alpha-\beta+(\lambda-1)\delta, \qquad f_Q=p+2-\lambda, \quad e_Q=(\lambda-1)\delta.$$

These are positive by (2.1). We claim that every term $Z^e b^j$ of f satisfies $j/f_P+e/e_P\ge 1$, that $f(0,b)$ has order exactly f_P, and that $f(Z,0)$ has order exactly e_P; and similarly for g with (f_Q,e_Q).

Indeed, $f(0,b)=A_l(1,b)$ has order $\rho_l=p+1-\lambda=f_P$ by Lemma 2.3. For a term coming from A_{l-t}: if $t\ge\lambda$ then $e=t\delta>e_P$; if $t<\lambda$ then $j\ge t-\lambda+N=f_P+t$, and $je_P+ef_P\ge(f_P+t)e_P+t\delta f_P>f_Pe_P$. For a term coming from B_{l-t}: if $t\ge\lambda-1$ then $e\ge e_P$; if $t<\lambda-1$ then $j\ge t+1-\lambda+N=f_P+t+1$, and

$$je_P+ef_P-f_Pe_P\ge(t+1)e_P+f_P(\alpha-\beta+t\delta)>0.$$

The constant term has $e=W_A>e_P$. Equality $j/f_P+e/e_P=1$ with $j=0$ occurs only for $B_{l-\lambda+1}$, whose index lies in $[1,m]$ and whose term free of b exists by Lemma 2.3, because $1-(l-\lambda+1)\equiv 0 \pmod N$; no term from an A_{l-t} has $e=e_P$ and $j=0$, since $j\equiv -1 \pmod N$ when $t\delta=e_P$. Its coefficient in U is free, so for generic U the order of $f(Z,0)$ is e_P. The same computation for g uses $g(0,b)=B_l(1,b)$ of order $\rho'_l=f_Q$. For B_{l-t} with $t<\lambda-1$ we have $e=t\delta$ and $j\ge f_Q+t$, so $je_Q+ef_Q-f_Qe_Q\ge te_Q+t\delta f_Q>0$; for A_{l-t} with $t<\lambda$ we have $e=\beta-\alpha+t\delta$, $j\ge f_Q+t-1$ and

$$je_Q+ef_Q-f_Qe_Q\ge f_Q(\beta-\alpha+t\delta)+(t-1)(\lambda-1)\delta>0;$$

the remaining terms have $e\ge e_Q$. The term Z^{e_Q} comes from $B_{l-\lambda+1}$ with a free coefficient in V, and as before no term from an A_{l-t} has $e=e_Q$ and $j=0$.

So the Newton polygon of f at the origin is the single segment from $(0,f_P)$ to $(e_P,0)$, and that of g is the segment from $(0,f_Q)$ to $(e_Q,0)$, in the (e,j)-plane (Figure 3). By the Weierstrass preparation theorem, f is a unit times a Weierstrass polynomial of degree f_P in b, and by the Newton–Puiseux theorem its roots $b_1(Z),\dots,b_{f_P}(Z)$ are Puiseux series of order $\sigma=e_P/f_P$. The intersection multiplicity is $\ell_a=\sum_i \operatorname{ord}_Z g(Z,b_i(Z))$ [5, Sections 1.2 and 2.6]. Now

$$f_Pe_Q<f_Qe_P,$$

because $f_Qe_P-f_Pe_Q=(\lambda-1)\delta+(p+2-\lambda)(\alpha-\beta)$, which is positive if $\alpha\ge\beta$, and if $\alpha<\beta$ it is at least $(\lambda-1)\delta-p(\beta-\alpha)$ with $\lambda\ge 2$, and $\delta-p(\beta-\alpha)=(pq-1-pq+p^2)/g>0$. Hence $\sigma>e_Q/f_Q$. Every term Z^eb^j of g contributes order $e+j\sigma\ge e+je_Q/f_Q\ge e_Q$ along each b_i, with equality only for the term Z^{e_Q}; so $\operatorname{ord}_Z g(Z,b_i(Z))=e_Q$ and $\ell_a=f_Pe_Q=(p+1-\lambda)(\lambda-1)\delta=(p-r)r\delta$.

The derivation D, and with it the whole construction, is unchanged when (a,p,A_l) and (b,q,B_l) are exchanged, and this exchange swaps F and G. Hence $\ell_b=(q-s)s\delta$. The finitely many genericity conditions used are open and nonempty. □

6. Proofs of the main results

Proof of Theorem 1. By Propositions 4.1 and 5.1,

$$\deg X_m=\frac{(\alpha+l\delta)(\beta+l\delta)}{\alpha\beta}-\frac{(p-r)r\delta}{\alpha}-\frac{(q-s)s\delta}{\beta}, \qquad l=m+1.$$

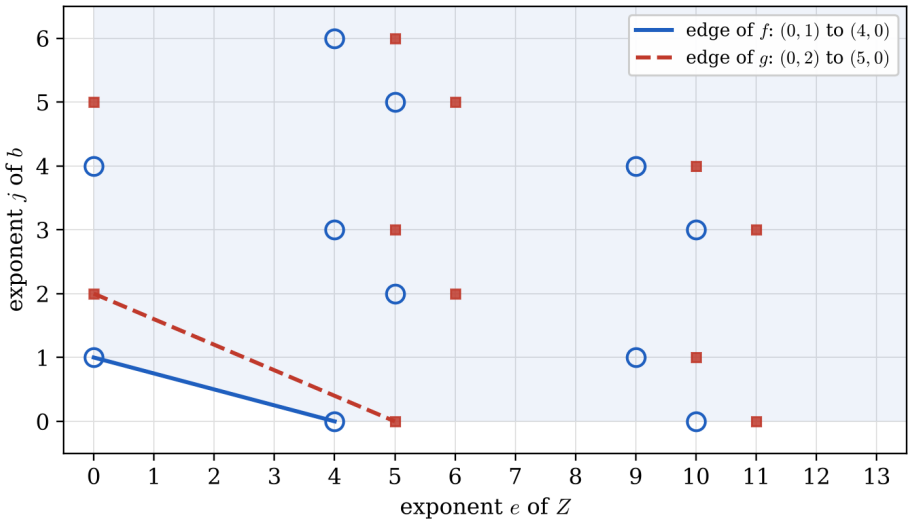


FIGURE 3. Terms $Z^e b^j$ of f (circles) and g (squares) at the point $[0:1:0]$ for $(p,q)=(2,3)$ and $m=4$, so that $\lambda=2$, $r=1$ and $\delta=5$. All terms lie on or above the edges from $(0,f_P)=(0,1)$ to $(e_P,0)=(4,0)$ and from $(0,f_Q)=(0,2)$ to $(e_Q,0)=(5,0)$, and the intersection multiplicity is $f_P e_Q = 5$.

Multiplying numerators and denominators by g and using $g\alpha = p+1$, $g\beta = q+1$, $g\delta = c$, we get $g(\alpha+l\delta) = p+1+(m+1)c = p(q+1)+mc$ and $g(\beta+l\delta) = q(p+1)+mc$, which is the formula. □

Proof of Corollary 2. For $p=q=n$, $c=n^2-1$, the first term is $(n(n+1)+m(n^2-1))^2/(n+1)^2 = ((n-1)m+n)^2$, and the bracket is $2(n-1)r(n-r)$ because $s=r$. For $n=2$ this is $(m+2)^2 - 2[\,r=1\,]$, and for $n=3$ it is $(2m+3)^2 - 4r(3-r)$, which is $(2m+3)^2-8$ for $r\in\{1,2\}$; this is (1.1). For $(p,q)=(2,3)$, $c=5$ and the formula becomes the one in (ii); evaluating it gives the listed values. Part (iii) follows from the shape of the formula, since $r(p-r)/(p+1)$ depends only on m mod $(p+1)$ and $s(q-s)/(q+1)$ only on m mod $(q+1)$. □

Remark 6.1. Theorem 3 and Section 3 hold for $p,q\geq 1$, while (2.1) fails for $p=1$: then $\alpha-\beta=-\delta$, the jet B_m has the same weight as A_{m+1}, and the highest weighted component of $A_{m+1}+U$ is $A_{m+1}+uB_m$ with a free coefficient u. The points at infinity are then the common zeros of $A_{m+1}+uB_m$ and B_{m+1}, and the method above needs an interlacing statement for this pencil. The degrees recomputed for the archive [15] agree with the formula of Theorem 1 also for $(p,q)=(1,2),(1,3),(1,5)$ and $m\leq 30$.

Remark 6.2. The proof gives more than the degree. For generic (U,V) the curves $A_{m+1}+U=0$ and $B_{m+1}+V=0$ in the weighted projective plane $\mathbb{P}(1,\alpha,\beta)$ [6] meet at infinity only at $[0:1:0]$ and $[0:0:1]$, with local ideals of the explicit lengths of Proposition 5.1, and all their affine intersection points are simple. Since $\mu^N/\kappa^M = -N/M$, Corollary 3.4 also shows that the zeros $z\neq 0$ of $A_l(1,z)$ are exactly the solutions of $z^{p+1} = \frac{p+1}{q+1}\cdot\frac{x}{x-1}$ with x a zero of F_l; in particular every such z^{p+1} is a negative real number.

DECLARATIONS

Data availability. No data were generated or analysed. Scripts that recompute the degrees of X_m for thirteen pairs (p,q) and $m\leq 30$, compare them with Theorem 1 and test Theorem 3

for small p, q, l are archived with the source of this paper on Zenodo, version 1.0.0, doi:10.5281/zenodo.24094790 [15]; no proof in the paper depends on them.

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